

**GANDER, NL, Canada**WMO: **717420**Lat: **48.946N**Lon: **54.567W**Elev: **495**StdP: **14.43**Time Zone: **-3.50 (NAF)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

| Coldest Month | Heating DB |     | Humidification DP/MCDB and HR |     |      |      |     |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-----|-------------------------------|-----|------|------|-----|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |     | 99.6%                         |     |      | 99%  |     |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99% | DP                            | HR  | MCDB | DP   | HR  | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c) | (d)                           | (e) | (f)  | (g)  | (h) | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 2             | 0.9        | 4.5 | -9.4                          | 3.4 | 2.8  | -5.0 | 4.3 | 5.8  | 39.5                  | 35.1 | 34.4 | 28.9 | 13.8                  | 280  | 0.767 |

**Annual Cooling, Dehumidification, and Enthalpy Design Conditions**

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 17.5                   | 79.6            | 65.7 | 76.8 | 64.4 | 73.9 | 63.2 | 69.0                | 75.8 | 67.1 | 73.3 | 65.3 | 71.1 | 12.1                 | 220  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 66.5                            | 99.4 | 72.7 | 64.7 | 93.3 | 70.7 | 62.9 | 87.5 | 68.8 | 33.4          | 75.9 | 32.0 | 73.2 | 30.5 | 71.2 | 75.0              |

**Extreme Annual Design Conditions**

| Extreme Annual WS |      |      | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |
|-------------------|------|------|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|
| 1%                | 2.5% | 5%   | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |
|                   |      |      | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  |
| (a)               | (b)  | (c)  | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |
| 29.8              | 26.3 | 23.3 | -4.0                       | 85.6 | 4.5                | 2.2 | -7.2                                               | 87.2 | -9.9       | 88.5 | -12.4      | 89.7 | -15.7      | 91.3 |
|                   |      |      | -4.9                       | 72.4 | 4.4                | 1.3 | -8.1                                               | 73.4 | -10.7      | 74.2 | -13.2      | 74.9 | -16.4      | 75.9 |
|                   |      |      | DB                         |      |                    |     |                                                    |      |            |      |            |      |            |      |
|                   |      |      | WB                         |      |                    |     |                                                    |      |            |      |            |      |            |      |

**Monthly Climatic Design Conditions**

|                                                                        |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |
|------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
|                                                                        |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 40.6   | 21.1  | 20.6  | 25.8  | 34.9  | 44.3  | 53.2  | 62.8  | 62.2  | 54.3  | 44.3  | 35.7  | 26.7  |
|                                                                        | DBStd       | 16.41  | 8.45  | 8.99  | 7.81  | 6.46  | 7.08  | 8.01  | 6.59  | 5.85  | 6.91  | 6.69  | 7.30  | 7.95  |
|                                                                        | HDD50       | 4585   | 896   | 822   | 750   | 455   | 210   | 56    | 3     | 2     | 33    | 203   | 433   | 722   |
|                                                                        | HDD65       | 8998   | 1361  | 1242  | 1215  | 904   | 642   | 361   | 119   | 122   | 325   | 640   | 880   | 1187  |
|                                                                        | CDD50       | 1158   | 0     | 0     | 0     | 1     | 32    | 152   | 398   | 381   | 163   | 28    | 3     | 0     |
|                                                                        | CDD65       | 98     | 0     | 0     | 0     | 0     | 0     | 7     | 49    | 36    | 6     | 0     | 0     | 0     |
|                                                                        | CDH74       | 603    | 0     | 0     | 0     | 0     | 10    | 71    | 318   | 176   | 28    | 0     | 0     | 0     |
|                                                                        | CDH80       | 70     | 0     | 0     | 0     | 0     | 1     | 7     | 42    | 18    | 2     | 0     | 0     | 0     |
|                                                                        | WSAvg       | 12.2   | 14.1  | 14.0  | 13.7  | 12.9  | 11.6  | 10.9  | 10.3  | 9.9   | 11.2  | 11.8  | 12.7  | 13.6  |
|                                                                        | PrecAvg     | 48.8   | 4.5   | 4.1   | 4.6   | 4.4   | 3.4   | 3.3   | 3.3   | 3.7   | 3.7   | 4.4   | 4.6   | 4.5   |
| (15) Precipitation                                                     | PrecMax     | 65.7   | 8.9   | 10.9  | 10.6  | 11.0  | 11.0  | 7.6   | 7.4   | 7.5   | 8.9   | 8.7   | 8.1   | 7.7   |
|                                                                        | PrecMin     | 27.2   | 1.5   | 1.4   | 0.9   | 1.6   | 0.0   | 1.0   | 0.4   | 0.5   | 0.0   | 0.8   | 0.2   | 1.6   |
|                                                                        | PrecStd     | 7.1    | 1.8   | 1.8   | 2.1   | 2.1   | 1.9   | 1.5   | 1.5   | 1.8   | 1.5   | 1.9   | 1.7   | 1.5   |
|                                                                        | PrecStd     | 7.1    | 1.8   | 1.8   | 2.1   | 2.1   | 1.9   | 1.5   | 1.5   | 1.8   | 1.5   | 1.9   | 1.7   | 1.5   |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB     | 49.9  | 47.5  | 47.7  | 60.1  | 73.8  | 80.1  | 84.3  | 82.3  | 77.9  | 66.9  | 59.4  | 51.3  |
|                                                                        |             | MCWB   | 48.9  | 45.3  | 43.2  | 50.5  | 58.5  | 64.3  | 68.6  | 68.5  | 65.8  | 61.3  | 56.5  | 50.2  |
|                                                                        | 2%          | DB     | 41.9  | 41.8  | 42.9  | 53.7  | 67.2  | 75.7  | 80.5  | 78.3  | 72.2  | 62.4  | 55.8  | 46.2  |
|                                                                        |             | MCWB   | 40.9  | 40.7  | 40.0  | 47.8  | 55.2  | 61.8  | 66.4  | 65.5  | 63.4  | 58.3  | 54.2  | 45.2  |
|                                                                        | 5%          | DB     | 37.1  | 36.9  | 39.2  | 49.1  | 61.9  | 71.7  | 77.6  | 75.6  | 68.8  | 58.9  | 51.9  | 41.2  |
|                                                                        |             | MCWB   | 36.0  | 35.4  | 36.3  | 43.7  | 52.4  | 60.2  | 65.1  | 64.5  | 61.7  | 55.1  | 50.3  | 40.2  |
|                                                                        | 10%         | DB     | 32.9  | 33.1  | 36.1  | 45.1  | 57.3  | 67.6  | 74.6  | 72.8  | 65.7  | 55.2  | 47.4  | 37.4  |
|                                                                        |             | MCWB   | 31.8  | 31.8  | 33.6  | 40.3  | 49.6  | 58.2  | 64.0  | 63.2  | 59.8  | 51.6  | 45.5  | 36.4  |
|                                                                        | 0.4%        | WB     | 48.8  | 46.4  | 45.3  | 53.6  | 61.3  | 68.6  | 71.7  | 70.8  | 68.4  | 63.2  | 57.3  | 50.4  |
|                                                                        |             | MCDB   | 49.7  | 47.3  | 47.7  | 56.2  | 68.6  | 75.8  | 80.0  | 78.2  | 73.6  | 65.3  | 58.5  | 51.0  |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 2%          | WB     | 41.2  | 40.7  | 40.3  | 48.5  | 57.5  | 65.0  | 69.5  | 68.4  | 65.7  | 59.2  | 54.2  | 45.3  |
|                                                                        |             | MCDB   | 41.9  | 41.7  | 41.9  | 52.9  | 64.6  | 71.3  | 76.9  | 74.7  | 70.2  | 61.4  | 55.6  | 46.0  |
|                                                                        | 5%          | WB     | 36.1  | 35.8  | 36.7  | 44.4  | 53.7  | 61.9  | 67.5  | 66.6  | 63.2  | 55.6  | 50.3  | 40.5  |
|                                                                        |             | MCDB   | 37.0  | 36.7  | 38.6  | 48.3  | 59.5  | 68.9  | 74.2  | 72.6  | 67.3  | 58.3  | 51.6  | 41.0  |
|                                                                        | 10%         | WB     | 32.0  | 31.9  | 34.0  | 40.9  | 50.5  | 59.1  | 65.6  | 64.9  | 60.8  | 51.9  | 45.7  | 36.5  |
|                                                                        |             | MCDB   | 32.6  | 32.7  | 35.8  | 44.3  | 56.9  | 66.2  | 71.7  | 70.7  | 64.7  | 54.7  | 47.2  | 37.3  |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR   | 12.6  | 13.3  | 13.0  | 13.2  | 16.9  | 18.1  | 17.5  | 15.9  | 15.0  | 12.0  | 10.7  | 10.4  |
|                                                                        |             | MCDBR  | 18.4  | 18.1  | 16.5  | 20.1  | 26.4  | 24.6  | 22.0  | 20.1  | 19.1  | 17.0  | 16.9  | 16.2  |
|                                                                        |             | MCWBR  | 18.9  | 17.9  | 14.3  | 14.3  | 16.1  | 14.2  | 11.6  | 10.7  | 12.0  | 13.8  | 16.6  | 16.5  |
|                                                                        | 5% WB       | MCDBR  | 18.3  | 18.0  | 15.5  | 18.8  | 23.6  | 21.4  | 19.3  | 17.6  | 17.2  | 16.5  | 16.9  | 16.2  |
|                                                                        |             | MCWBR  | 18.8  | 18.2  | 14.6  | 15.1  | 16.1  | 13.8  | 11.7  | 11.2  | 12.7  | 14.9  | 16.9  | 16.7  |
|                                                                        |             | MCWBR  | 18.8  | 18.2  | 14.6  | 15.1  | 16.1  | 13.8  | 11.7  | 11.2  | 12.7  | 14.9  | 16.9  | 16.7  |
| (40) Clear-Sky Solar Irradiance                                        | taub        |        | 0.304 | 0.302 | 0.332 | 0.351 | 0.366 | 0.388 | 0.407 | 0.388 | 0.355 | 0.332 | 0.309 | 0.302 |
|                                                                        | taud        |        | 2.274 | 2.342 | 2.324 | 2.385 | 2.404 | 2.358 | 2.347 | 2.397 | 2.482 | 2.511 | 2.496 | 2.325 |
|                                                                        | Ebn at Noon |        | 235   | 270   | 279   | 285   | 284   | 277   | 269   | 270   | 269   | 257   | 239   | 220   |
|                                                                        | Edh at Noon |        | 23    | 28    | 33    | 35    | 36    | 37    | 37    | 34    | 29    | 23    | 19    | 20    |
| (44) All-Sky Solar Radiation                                           | RadAvg      |        | 306   | 548   | 908   | 1254  | 1486  | 1589  | 1647  | 1357  | 1039  | 635   | 347   | 226   |
|                                                                        | RadStd      |        | 27    | 65    | 82    | 142   | 119   | 180   | 135   | 100   | 73    | 67    | 38    | 21    |

**Historical Trends**

|      |                         | DBAvg | Heating |        |       | Cooling |       |     | Degree-Days |       |       |       |
|------|-------------------------|-------|---------|--------|-------|---------|-------|-----|-------------|-------|-------|-------|
|      |                         |       | 99% DB  | 99% DP | 1% DB | 1% WB   | 1% DP |     | HDD50       | HDD65 | CDD50 | CDD65 |
|      |                         | (d)   | (e)     | (f)    | (g)   | (h)     | (i)   | (j) | (k)         | (l)   | (m)   |       |
| (46) | Station Only            | N/A   | N/A     | N/A    | N/A   | +0.77   | +1.28 | N/A | N/A         | N/A   | N/A   |       |
| (47) | Regional (12 neighbors) | N/A   | N/A     | N/A    | N/A   | N/A     | +0.95 | N/A | N/A         | N/A   | N/A   |       |

Nomenclature: See separate page